higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CHEMISTRY
NQF LEVEL 4

13 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 False
1.2 True
1.3 False
1.4 True
1.5 True

(5 × 1) **[5]**

QUESTION 2

2.1 D
2.2 E
2.3 B
2.4 A
2.5 C

(5 × 1) **[5]**

QUESTION 3

3.1 Catalyst
3.2 Floatation process
3.3 Gasification
3.4 Freezing point
3.5 Mould/Fungi

(5 × 1) **[5]**

QUESTION 4

4.1 4.1.1
- External lubricants
- Internal lubricants (2)

4.1.2
- They help the flow of melted polymers/plastic by reducing viscosity (surface tension).
- They give the end product a glossy surface.
  They prevent hot melts from sticking to the machine surface/processing equipment.
- They help in the dispersion of pigments and fillers in the polymers being produced. (Any 3 × 1) (3)

4.2 A VCM (vinyl chloride monomer)
B Water
C Suspending agent
D Polymerisation reactor
E Centrifuge
F Dry PVC powder (6)

4.3 4.3.1 Stripping process (1)

4.3.2 • To remove/reduce water from the slurry
• To separate light and heavy components (Any ONE) (1)

4.3.3 To start the polymerisation process in polymer production/ synthesis (1)

4.3.4 Hot dry air (1)

4.4 •Chlorine
•Carbon
•Hydrogen (3)

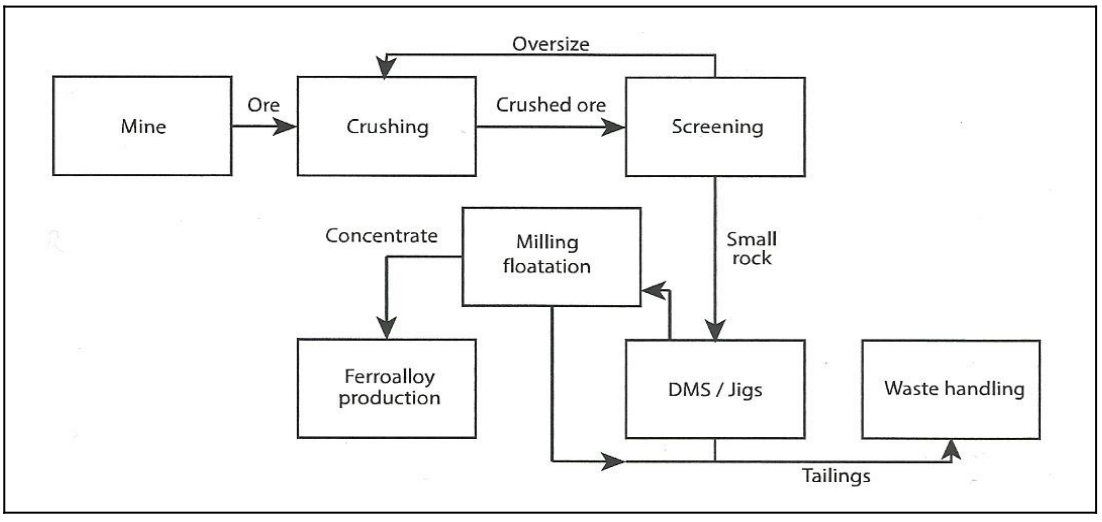
4.5 •Rubber
•Elastomers
•Thermoplastics resins
•Thermo-hardening resins (4)

4.6 •Injection moulding
•Blow moulding
•Extrusion
•Rotational moulding
•Compression moulding (5)
[27]

QUESTION 5

5.1 •It is used in the production of electronics, including computers, mobile phones and televisions
•It is used in dentistry
•It is used in ground water treatment
•It is used in catalytic converters in diesel cars and trucks (4)

5.2



(13)

5.3
- Roossenekal
- Magnet Heights
- Pretoria
- Brits
- Rustenburg
- North of Pilanesberg (5)

[22]

QUESTION 6

6.1 6.1.1 Dimethyl benzene/1,2-dimethylbenzene/1,3-dimethylbenzene/1,4-dimethylbenzene (1)

6.1.2
- Meta-xylene (M-xylene)
- Ortho-xylene (O-xylene)
- Para-xylene (P-xylene) (3)

6.1.3
- Blending component in car fuels
- Used as a solvent in paints and printing ink, adhesives, perfumes, pharmaceuticals and films (2)

6.2
- Coke
- Light gas
- Tar
- Light oil
- Ammonia (5)

6.3 Phenol mixture is first distilled in the solvent recovery unit✓ to separate the phenols from the solvent.✓ Steam stripping✓ of the phenols is then used to recover any remaining solvent from phenols.✓ The solvent is returned to the extraction section where it will again be used to extract phenols.✓ (5)

6.4 6.4.1 Paraffin (1)

6.4.2
- Used for lamps/lighting
- Used for aviation (jet fuel)
- Used for cooking
- Used for heating (Any 3 × 1) (3)

6.5
- Separation
- Conversion
- Heat treatment (3)

[23]

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QUESTION 7

7.1
- Barley/Sorghum
- Water
- Yeast
- Hops (4)

7.2 Barley grains are soaked in water to absorb about 45% water✓ and to activate the metabolic process of the dormant kernel.✓ During this process, water is drained✓ and the grains turned several times to increase oxygen uptake by barley.✓ (4)

7.3
- Nutrient medium preparation ✓
- Heat sterilisation of the nutrient medium and propagation vessel (inoculum growing equipment) ✓
- Preparation of the seed culture/propagation (growing) in the sterile nutrient medium in sterile propagation vessel (equipment) ✓ (3)

7.4 After fermentation at optimum temperature, the aqueous liquid (broth) containing penicillin is rapidly cooled to 5 °C ✓using appropriate heat exchangers to reduce/minimise penicillin degradation during the extraction step. ✓ (2)

[13]

TOTAL: 100

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